

Amr Farahat

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Education

PhD	International Max Planck Research School for Neural Circuits, Donders Centre for Neuroscience, Radboud University	Frankfurt, Germany, Nijmegen, the Netherlands 11/2020 – 10/2025
	Thesis: On the predictive and explanative roles of deep neural networks in neuroscience.	
MSc	Otto von Guericke University , Integrative Neurosciences	Magdeburg, Germany 2015 – 2018
	- Thesis: Deep Learning for EEG Decoding and Automatic Feature Discovery. - Grade: with distinction (1.2)	
MD	Mansoura University , Medical School	Mansoura, Egypt 2007 - 2014

Experience

Green Fusion , Data Science Engineer	Berlin, Germany 12/2025 - Present
Ernst Strüngmann Institute for Neuroscience in Cooperation with Max Planck Society , PhD researcher	Frankfurt, Germany 11/2020 - 10/2025
Max Planck Institute for Brain Research, Frankfurt Institute for Advanced Studies , Predoctoral researcher	Frankfurt, Germany 11/2019 - 10/2020
Otto von Guericke University Hospital , Neuroradiology Resident Doctor	Magdeburg, Germany 4/2019 - 10/2019
Ministry Of Health and Population , General Practitioner	Ras Ghareb, Egypt 2015

Projects

Deep Learning for EEG Decoding and Automatic Feature Discovery.	Publication Abstract Code
- Developed CNN models for classifying P300 ERP component in EEG data for a Brain Computer Interface (BCI) speller application. - Used saliency maps to extract the relevant spatial and temporal EEG features that drive the model and reflect the neural correlates of covert attention. - Tools Used: Python, Tensorflow, EEG data	
Diagnosing Epileptogenesis with Deep Anomaly Detection.	Video presentation Publication Abstract Code
- Used data from a rodent epilepsy model to show the feasibility of using an unsupervised deep anomaly detection framework using adversarial autoencoders to detect subtle changes in brain electrical activity triggered by brain injury that lead eventually to an epileptic brain. - Tools Used: Python, Tensorflow, intracranial EEG data	
A novel feature-scrambling approach reveals the capacity of convolutional neural networks to learn spatial relations.	Publication Abstract
- Developed a feature-scrambling approach in order to investigate the granularity of features used by CNNs for object recognition and whether CNNs are capable of encoding the spatial relations among features to build more complex diagnostic features for objects. - Tools Used: Python, Tensorflow	

Neural responses in early, but not late, visual cortex are well predicted by random-weight CNNs with sufficient model complexity.

[Publication](#) 

- Evaluated how well both untrained (random-weight) and trained CNNs predict neural activity across the visual cortex in both humans and monkeys, systematically varying architectural components like activation functions and pooling methods.
- Tools Used: Python, Tensorflow, fMRI data, Electrophysiology data

Investigating motor learning circuitry in a 7T functional connectivity fMRI study.

[Poster](#) 

- Evaluated how well both untrained (random-weight) and trained CNNs predict neural activity across the visual cortex in both humans and monkeys, systematically varying architectural components like activation functions and pooling methods.
- Tools Used: Python, Matlab, fMRI data, Presentation software

Selected Conference Presentations

The role of spatial relations among CNN features of different granularities in object recognition. Bernstein Conference	Berlin, Germany 2022
How do convolutional neural networks understand shape? Champalimaud Research Symposium	Lisbon, Portugal 2021
Implicit motor learning modulates functional connectivity of DBS therapeutic target for tremor. Neurorad Conference	Frankfurt, Germany 2019
Convolutional neural networks for EEG decoding and exploration of brain dynamics. Bernstein Conference	Berlin, Germany 2018

Further Qualifications

The Machine Learning Summer School in Okinawa	Okinawa, Japan 2024
The Systems Vision Science Summer School & Symposium	Tübingen, Germany 2023
IBRO-Simons Computational Neuroscience Imbizo	Cape Town, South Africa 2022
FENS Summer School on ‘Artificial and natural computations for sensory perception: what is the link?’	Bertinoro, Italy 2022
International Startup School. Transfer- und Gründerzentrum (TUGZ), Otto von Guericke University	Magdeburg, Germany 2018

Skills

Technical skills: Python, Matlab, EEG, fMRI, Tensorflow, computer vision
Languages: English (TOEFL 109), German (C1), Arabic (native)

Publications

- [1] A. Farahat and M. Vinck, “Neural responses in early, but not late, visual cortex are well predicted by random-weight cnns with sufficient model complexity,” *bioRxiv*, pp. 2025–02, 2025.
- [2] A. Voegtler, L. Terzic, A. Farahat, *et al.*, “Ventrointermediate thalamic stimulation improves motor learning in humans,” *Communications Biology*, vol. 7, no. 1, p. 798, 2024.
- [3] A. Farahat, F. Effenberger, and M. Vinck, “A novel feature-scrambling approach reveals the capacity of convolutional neural networks to learn spatial relations,” *Neural Networks*, vol. 167, pp. 400–414, 2023.
- [4] A. Farahat, D. Lu, S. Bauer, *et al.*, “Diagnosing epileptogenesis with deep anomaly detection,” *Proceedings of the 7th Machine Learning for Healthcare Conference*, vol. 182, 2022.

- [5] L. Terzic, A. Voegtle, A. Farahat, *et al.*, "Deep brain stimulation of the ventrointermediate nucleus of the thalamus to treat essential tremor improves motor sequence learning," *Human Brain Mapping*, 2022.
- [6] A. Voegtle, C. Terlutter, K. Nikolai, A. Farahat, H. Hinrichs, and C. M. Sweeney-Reed, "Suppression of motor sequence learning and execution through anodal cerebellar transcranial electrical stimulation," *The Cerebellum*, pp. 1–14, 2022.
- [7] A. Farahat, C. Reichert, C. M. Sweeney-Reed, and H. Hinrichs, "Convolutional neural networks for decoding of covert attention focus and saliency maps for eeg feature visualization," *Journal of neural engineering*, vol. 16, no. 6, 2019.